SYSTEM DESIGN

V 1

The Problem

Current Challenges in Recycling:

- Electronic waste is rapidly increasing, with millions of devices discarded annually.
- Manual disassembly of electronic devices is labor-intensive and time-consuming.
- Workers lack systematic guidance on how to safely and efficiently take apart different types of devices.

Need for a Systematic Approach:

- Disassembling new types of objects requires detailed instructions.
- Ensuring worker safety and adherence to environmental regulations is critical.
- Collecting data on disassembly processes can optimize and improve efficiency.

Objective of the System:

- To develop a data collection system that captures detailed instructions and steps for disassembling various electronic devices.
- To provide a structured guide for workers, improving safety, efficiency, and accuracy in the disassembly line.
- To gather valuable data that can be analyzed to enhance disassembly processes and workflows.

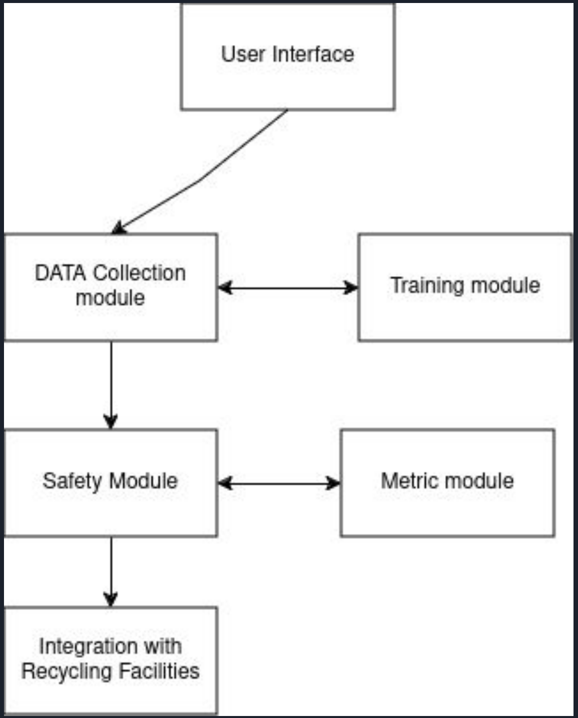
Introduction to the Proposed System:

- A data collection system designed to capture detailed instructions and steps for disassembling various electronic devices.
- Provides a structured guide for workers to improve safety, efficiency, and accuracy in the disassembly line.
- Gathers valuable data that can be analyzed to enhance disassembly processes and workflows.

Key Components of the System:

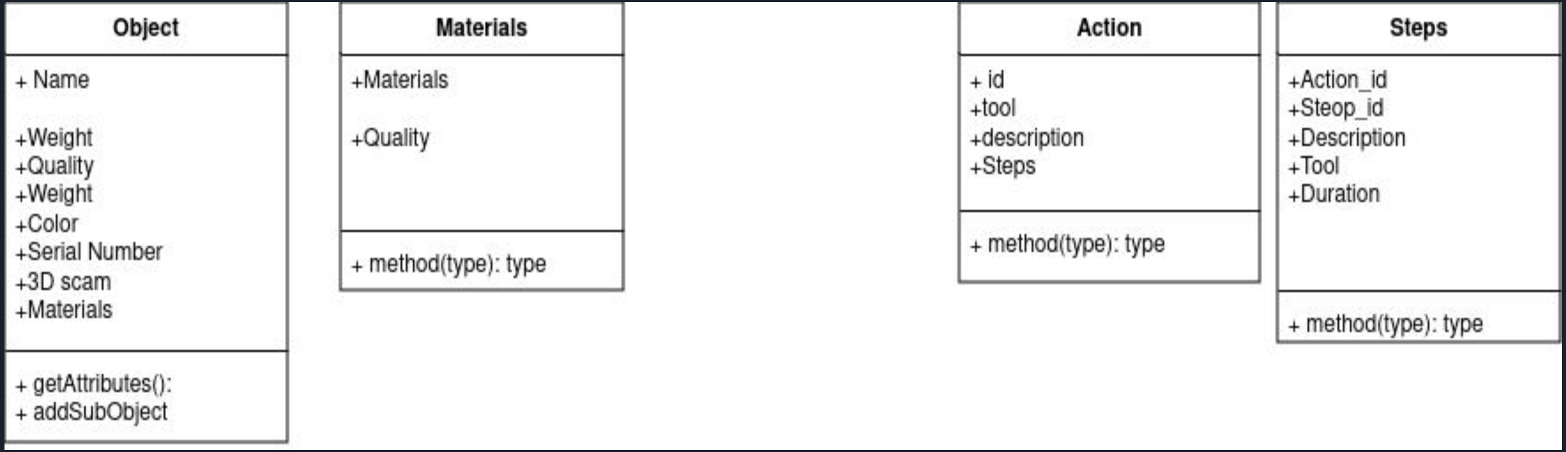
- **Data Collection Module:** Captures detailed information about each object and its disassembly steps.
- **Worker Interface:** A user-friendly interface that guides workers through the disassembly process step-by-step.
- **Training Module:** Provides training materials and assessments to ensure workers are well-prepared.
- **Safety Protocols:** Ensures that all disassembly actions adhere to safety regulations and guidelines.
- **Performance Metrics and Reporting:** Tracks and reports the efficiency and effectiveness of the disassembly process.
- **Integration with Recycling Facilities:** Connects with existing recycling systems for streamlined operations.

High-Level Diagram or Flowchart



- **User Interface**:
  - Central point where workers interact with the system.
- **Data Collection Module**:
  - Captures detailed information about objects and disassembly steps.
- **Training Module:**
  - Offers training resources and assessments for workers.
- **Safety Protocols Module**:
  - Ensures all actions adhere to safety regulations.
- **Performance Metrics & Reporting**:
  - Provides reports for analysis and improvement.
- **Integration with Recycling Facilities**:
  - Connects with external recycling systems for seamless operations.

UML Class Diagram



System Requirements

Data Collection Module:

- The system must allow users to enter detailed information about new objects and their components.
- The system must store disassembly steps for each object in a database.
- The system must provide an interface for updating and managing disassembly instructions.

User Interface:

- The system must provide a user-friendly interface for workers to follow disassembly instructions.
- The interface must be interactive and guide workers step-by-step through the disassembly process.
- The system must support multilingual options to accommodate a diverse workforce.

Training Module:

- The system must include training materials and tutorials for new workers.
- The system must conduct assessments to evaluate worker readiness.
- The system must provide feedback and additional training resources based on assessment results.

Safety Protocols:

- The system must include safety guidelines and protocols for each disassembly action.
- The system must perform real-time safety checks during the disassembly process.

Performance Metrics and Reporting:

- The system must provide tools for analyzing performance data to identify areas for improvement.

Integration with Recycling Facilities:

- The system must integrate with external recycling systems through APIs.
- The system must manage logistics for transporting disassembled parts to recycling facilities.